

RICHLANDS ELLIS, NC, USA

WMO: 723069

Lat: 34.833N

Lon: 77.617W

Elev: 96

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93753

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	24.7	-8.5	3.5	41.7	2.7	6.3	40.4	23.8	62.9	20.4	60.5	2.6	20	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	94.6	77.5	92.0	76.7	90.3	76.0	80.3	90.5	79.0	88.9	77.9	86.9	8.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	143.1	84.6	76.2	137.5	83.5	75.2	132.5	82.4	43.7	90.5	42.5	88.9	41.3	87.2	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.5	17.1	15.2		DB	12.8	99.1	4.1	2.5	9.9	100.8	7.5	102.3	5.3	103.7	2.3	105.5
				WB	10.9	81.9	4.4	2.1	7.8	83.4	5.2	84.6	2.7	85.8	-0.5	87.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	43.6	46.6	52.5	61.3	69.4	76.5	79.9	78.5	73.2	63.3	52.6	46.9
	DBStd	15.03	10.77	9.58	9.36	8.26	6.94	5.19	3.81	4.13	5.50	8.11	8.98	9.90
	HDD50	776	258	164	84	9	0	0	0	0	0	4	74	183
	HDD65	2876	665	519	398	166	38	2	0	0	8	131	383	566
	CDD50	5199	60	69	163	348	602	796	926	884	696	418	151	86
	CDD65	1825	2	4	12	55	176	348	461	419	254	80	10	4
	CDH74	16170	4	25	128	566	1649	3215	4396	3648	1875	596	55	13
	CDH80	6419	0	1	15	139	581	1384	1996	1545	629	125	4	0
(14) Wind	WSAvg	6.2	6.1	6.6	7.4	7.8	6.8	6.0	5.8	5.1	5.7	5.4	5.5	5.7
(15) Precipitation	PrecAvg	55.6	3.9	3.6	4.1	3.1	4.2	5.3	6.5	7.0	6.7	4.0	3.4	3.6
	PrecMax	73.9	7.8	10.1	8.2	6.6	8.2	12.4	12.0	14.6	21.7	11.0	8.5	7.6
	PrecMin	35.7	1.3	1.4	1.5	0.5	1.9	1.9	3.7	2.3	0.9	0.1	0.7	0.8
	PrecStd	9.3	1.6	1.9	1.7	1.5	1.7	2.4	2.2	2.8	4.6	2.8	2.0	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	78.5	81.9	88.1	93.5	97.2	98.6	97.1	91.6	87.9	79.3	75.2
		MCWB	63.9	66.2	64.3	67.7	73.5	76.5	79.2	79.2	76.0	73.2	67.4	65.7
	2%	DB	69.9	72.2	77.8	83.8	90.0	93.1	94.8	92.9	89.7	82.6	75.0	71.9
		MCWB	61.9	61.3	64.2	66.9	71.9	76.0	78.7	77.8	75.5	71.0	65.3	63.9
	5%	DB	65.7	67.9	73.2	81.0	86.3	90.6	92.1	90.7	86.5	80.8	72.0	67.7
		MCWB	59.3	59.9	60.9	65.3	71.1	75.3	77.8	77.2	74.3	69.7	63.7	62.6
	10%	DB	62.0	63.8	69.8	76.9	82.4	88.2	90.1	88.3	83.7	77.1	68.2	63.8
		MCWB	55.1	56.2	59.4	63.6	69.0	74.3	77.0	76.3	73.1	68.2	62.0	58.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.7	67.7	68.7	73.8	76.5	80.2	82.4	81.7	78.9	77.0	71.4	69.1
		MCDB	70.8	75.0	76.7	80.2	86.2	91.4	93.6	91.5	86.9	82.7	76.0	73.9
	2%	WB	63.9	64.8	66.2	70.4	74.7	78.4	80.6	80.0	77.4	74.7	68.2	66.2
		MCDB	67.7	69.5	73.7	78.5	84.6	89.2	91.1	89.9	85.4	79.7	72.7	69.7
	5%	WB	60.7	61.5	63.9	68.1	73.1	77.1	79.4	78.7	76.1	72.8	65.7	63.4
		MCDB	64.9	66.0	69.8	76.6	82.7	87.1	89.7	87.8	83.3	77.6	69.9	66.3
	10%	WB	56.7	57.5	61.6	66.0	71.4	75.7	78.1	77.5	74.7	70.2	62.8	59.7
		MCDB	60.8	63.4	68.1	74.3	80.2	84.6	87.3	85.6	81.2	75.4	67.1	63.6
(35) Mean Daily Temperature Range	MDBR		20.9	22.3	24.1	24.3	22.8	20.6	19.5	18.7	18.9	21.7	23.0	20.9
	5% DB	MCDBR	25.0	27.0	28.0	27.8	25.5	23.5	22.0	21.6	21.9	23.0	24.8	22.5
		MCWBR	18.6	18.3	16.2	14.5	11.6	9.8	9.2	9.2	10.0	12.7	16.3	16.6
	5% WB	MCDBR	21.9	24.5	23.0	23.2	21.2	21.0	20.0	19.2	18.9	17.3	20.1	20.9
		MCWBR	18.6	19.1	15.7	13.8	10.3	9.6	8.9	8.6	9.6	10.3	15.8	17.4
(40) Clear-Sky Solar Irradiance	taub		0.318	0.330	0.359	0.397	0.452	0.492	0.525	0.506	0.421	0.375	0.339	0.323
	taud		2.518	2.479	2.422	2.311	2.201	2.134	2.057	2.105	2.343	2.446	2.502	2.539
	Ebn at Noon		282	291	290	282	267	255	246	248	266	271	272	273
	Edh at Noon		26	30	35	41	46	49	53	49	37	31	26	24
(44) All-Sky Solar Radiation	RadAvg		844	1070	1410	1771	1941	2022	1919	1737	1448	1219	933	747
	RadStd		53	103	105	112	137	114	110	102	102	139	69	57

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+286	+207

Nomenclature: See separate page